Cryptocurrency Market Order Book Prediction Using Deep Learning Methods

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Project Summary

The cryptocurrency markets have exploded in popularity in recent years, with the price of Bitcoin (BTC) increasing over 1000% in just the last 5 years. For much of its history, institutional investors did not take BTC seriously and treated it as a fringe asset essentially equated with gambling. Large financial institutions have now developed exchange-traded funds (ETFs) to package and sell exposure to BTC to the broader market. Additionally, companies like Microstrategy have emerged, which use BTC as a primary means of structuring their capital. These factors underscore the critical importance of the problem of BTC price level forecasting and analytics.

This problem is similar to the well-studied problem of stock price prediction and thus presents many of the same challenges. One such challenge is the low signal-to-noise ratio, which refers to the difficulty in distinguishing meaningful market trends from random price fluctuations. Another challenge is the limited availability and granularity of pricing data, which means that the data used for analysis may not be comprehensive or detailed enough. These factors make BTC price forecasting a complex task, especially when considering applying Deep Learning methods, which can have problems with overfitting to noise and typically requires large amounts of data.

However, the increasing availability and access to rich order book-level data due to the growing prevalence of so-called high-frequency trading (HFT) provides an ideal avenue to apply these methods. Recent research has shown that high-resolution order book-level data provides information about market activity to capture intricate market dynamics and provide increasingly accurate forecasts.

Our research aims to leverage current state-of-the-art deep learning approaches to address this important problem. We seek to couple this with domain expertise provided by our team members to bridge the commonly observed gap between purely academic research and practical implementations that could inform strategies that can actually be monetized via trading strategies.

Approach

We will implement and train linear, CNN, transformer, and CNN-transformer hybrid models on Bitcoin order book data, compare the predictive performance of deep learning approaches against traditional methods. Initially, we will treat the task as a classification problem by labeling price movements as upward, downward, or stationary before transitioning to a regression-based approach to predict exact price changes. Our study leverages key datasets and methodologies, including the deep-orderbook repository for Bitcoin order book tick data (1), the **TLOB model** for transformer-based price prediction (2), and an open-source package providing end-to-end components for data processing and model training (3). To ensure a rigorous evaluation, we will apply hyperparameter tuning and assess model performance using both classification and regression metrics. Furthermore, we will implement back-testing and strategy-driven evaluation techniques to measure the real-world applicability of our models in predicting trade entry and exit points. If time permits, we will expand our analysis to additional cryptocurrency markets to test the generalizability of our approach.

Resources / Related Work & Papers

The advent of electronic trading, and especially the use of the electronic Limit Order Book (LOB) has facilitated an explosion of quantitative and computational trading algorithms. More and more, computers are used to generate trade signals, especially in the realm of high-frequency trading. Traditional trading algorithms often relied on stochastic methods but more and more, financial practitioners have recognized that the vast quantities of financial data are a valuable resource that can be fed into Machine Learning models to uncover patterns that humans cannot discern on their own. The financial trading problem is far from solved, however, as significant challenges such as high signal-to-noise ration, non-stationarity, market volatility and the complex effects of the socio-polito-economic environment prevent are constantly evolving the market. Moreover, due to the efficient market hypothesis, any pattern that is uncovered and exploited gets quickly arbitraged away and so does not remain a viable trading signal for long.

Following the enthusiastic adoption of Deep Learning in the Computer Vision community, Deep Learning was quickly recognized as a powerful new tool that could aid financial practitioners in analyzing financial data for trading to search for non-linear, non-intuitive, and deeply embedded and abstract relationships in the data. As early as 2017, LSTMs and CNNs were proposed for the purposes of stock price prediction with good results (7). These two architectures proved to be extremely robust and high performing for quite some time. A number of other architectures were proposed soon after, usually just different variations and tweaks of these two core architectures. Eventually they were combined together in DEEPLOB (8), which allowed leveraging the different advantages and biases of the two networks together for even better performance. CNNs were useful as they allowed learning of abstract higher-order features of the LOB, and LSTM layers allowed learning the temporal dynamics in the time-series. Attention was soon added to create DEEPLOBATT (9). Naturally, as the Transformer model soon came to pass RNNs like LSMTs in the wider Deep Learning community, it also became the more popular architecture in the financial time-series prediction research. MASTER was a model that used multiple Transformer attention layers to capture both inter-stock and intra-stock and market information and decide which features to pay attention to in order to achieve superior performance (10). Recently, TLOB was one that applied the Transformer architecture on the LOB itself (2).

These deep learning models have been applied across a variety of different asset classes, including both equitites and crypto, at multiple time frequencies. However, due to the incredibly complex and noisy information environment, regime shifting and the inherent non-stationarity of financial time-series data, performance of one model does not necessarily translate to other asset classes, time-frequencies, or even different time-ranges. As with all of Machine Learning, the data determines everything, and if the inherent properties of the data change, so too will model performance. Just because Transformer models have been getting a lot of popularity in recent literature does not mean they necessarily outperform. A recent review of deep learning models on daily crypto data (11) showed that CNN-LSTM hybrid approaches still outperformed all models including Transformers. However, this review was performed on daily data during the COVID-19 pandemic. It is unclear, for example, how the different models compare on higher-frequency LOB data at a more current time range.

Our project aims to help improve the state of understanding of how the latest state-of-the-art models, such as TLOB, perform and compare against each other on higher-frequency crypto limit-order-book data. This is a data regime that many of the state-of-the-art models have not been applied to as they focus on either LOB data for equitites or daily crypto data. This research will shed further light on how the different architectures, such as TLOB, are affected by different data environments and whether or not they generalize to domains different than the ones that they were originally designed for.

Datasets

High Frequency Limit Order Book: The limit order book data were from the cryptocurrency exchange coinbase. The data span 9 days with 50 levels of non-zero best bid and best ask, contains snapshots every 100ms up to a depth of 50 for 9 consecutive days, from the 12th to the 20th of June 2019. Each order is described by its price and volume. A depth of 50 for both bid and ask at each timestep contains 200 features. This data set is open to the public at Globe Research's Github.

High Frequency Crypto Limit Order Book Data, The dataset was created from the raw order book and aggregated to construct snapshots of the limit order book at frequencies of 1 second, 1 minute, and

5 minutes. And the dataset contains roughly 12 days of limit order book data for Bitcoin (BTC), Ethereum (ETH) and Cardano (ADA). The data contains information for the 15 best bid / ask price levels in the order book. The data in 9 csv files is open to the public at Kaggle .

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